

Vaccine Inventory Management Using Receding Horizon Method

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Abstract— For the 2100 pandemic, this paper simulates the cost of vaccine administration in a 52-day warehouse. Based on the optimal fixed level storage $r = 1$ found based on the Monte Carlo method, the minimum warehouse management cost of variation under dynamic random inoculation headcount is modelled using the receding horizon method with varying order numbers y . Finally, the inability to converge to near-optimal costs due to the randomness of the arrivals is eventually solved by introducing stochastic model prediction (SMPC).

I. INTRODUCTION

The mission issued by the United Nations is to minimize the cost of storing vaccines in warehouse in the 2100 pandemic. The optimal fixed level of storage $r = 1$ can be obtained easily if the fixed order strategy is adopted and the Monte Carlo method [1] is used. The cost overhead of the warehouse consists of three components: a storage fee of 5 gold coins per unit, a penalty fee of 20 gold coins per day for the number of vaccines less than the number of vaccinators, and a clearance fee of 10 gold coins per unit for the last day of storage. The difficulty of the problem is that the order y per day is not fixed, and the resulting huge computation makes it almost impossible to rely on Monte Carlo to predict the minimum cost in this model. Therefore the receding horizon method [2] of updating the state according to the actual dynamics of the system is considered. In this method it is necessary to approach the optimal convergence state by minimising the cost function. However it is difficult to write a cost function directly because of the complex relationship between the remaining stocks and the order y , the number of random arrivals per day and the final cost to be calculated. As the cost function ultimately needs to be minimised in the receding horizon method, it is equated here to the cost, as the cost also needs to be minimised in this task. By predicting the cost for a fixed number of days in the future, find the number of incoming per day order y that minimises it, take a fixed control number of days of y and apply it to the actual arrivals, and then roll forward through the process based on the remaining stock obtained in the present moment. Unfortunately, since the number of arrivals for this task is uniformly distributed with a certain probability, the randomness is too large, which ultimately prevents the results from converging. To address this problem, the model is improved to SMPC [3], which deals with the uncertainty by introducing a probabilistic model, and the final result is converged.

II. PROCESS OF COMPLETING THE MISSION

A. Choosing y

Since the order number y is not fixed for per day, to minimise warehouse cost, the order number y on the first day should be equal to the number of people arriving on the next day, namely there are no remaining vaccines that need to be stored everyday,

and there are no penalties for not having enough vaccines. However, since the number of arrivals per day is uncorrelated, it is not possible to predict the exact number of arrivals on the second day on the first day, so the final optimal order number y should approximate the probability distribution of arrivals.

B. Receding Horizon Method

Based on the background the following system can be constructed:

$$x_{k+1} = f_k(x_k, u_k) - w_{k+1}$$

$$y_k = h_k(x_k, u_k)$$

[2] And the x_k is the state which means the remaining stocks in k day, the u_k is the order y in k day, and the w_{k+1} [4] is number of arrivals in $k + 1$ day. Because the w_{k+1} is a uniform distribution based on a certain probability, so w_{k+1} is the disturbance in the system. And y_k is the cost in k day. Because whether or not to order is related to the number of the remaining stocks, so this system is not easy to write in standard form. And for this optimal design, the performance index is:

$$J = \sum_{k=k_0}^{k_0+N-1} l(x_k, u_k)$$

Where $1 \leq u_k \leq 6$, and N is the length of the prediction horizon. Obviously in this task J stands for minimum cost. Choosing a prediction horizon of 10 days, namely $N = 10$, and a control horizon of 5 days, and applying the receding horizon method of traversing one hundred rounds, the minimum cost obtained is 390 at the minimum and 660 at the maximum. Obviously, this result is rather unstable, but it still has advantages over the Monte Carlo method. First of all it is a dynamic result, namely the optimal daily order is related to the current number of arrivals in the actual system, which Monte Carlo cannot do. Secondly the computational effort is very small compared to Monte Carlo. Finally, since a variable order number y is used, it is possible to end up with a lower cost than with a fixed order number y .

C. Changing the control and prediction horizons

In order to allow the system to change the control horizon length and the predicted horizon length freedly, an adaptive control method is thus designed. That is, it is obvious that the last day should be considered separately, so that the actual predicted horizon length is 51 days at most. Assuming that the

length of the control horizon is b days, the number of rounds for which predictions need to be made is:

$$l = b \parallel 51$$

The number of days remaining is:

$$d = (51 \bmod b) + 1$$

That is, for the first l rounds, the set control horizon and prediction horizon are applied. For the remaining d days, a one-step prediction and one-step control is applied. And then keep the prediction horizon to 10, change the control horizon from 1 to 5, the result is as the length of the control horizon increases, the predictions become more unstable. This is because if the number of steps in the control horizon increases, the behaviour of making changes to the state in the actual system decreases, thus increasing the noise disturbance. And keep the control horizon to 1, change the prediction horizon from 2 to 5, the predictions also become more unstable. This is because as the prediction horizon increases, the disturbance in the prediction system become cumulatively larger and therefore the results are more unstable.

D. SMPC

It is clear from the above discussion that while RHC [2] can make the best judgement of the moment, the results are not always stable. This is due to the fact that the disturbance w_{k+1} [4] is too random, leading to the possibility that the predicted system may be too far from the actual system. However, for this task it is not possible to get the actual number of arrivals on the next day. The improvement is centred on two points. Firstly, because the number of days 52 is not very long, the actual arrival system based on probabilistic simulation may not be accurate. A more accurate actual random system should be generated first. Second, the predicted stochastic system may also be inaccurate. However, since the predicted number of days is much shorter, it is not possible to judge whether it is correct or not by its probability, but only whether it is possible from its probability. In fact, the optimal number of orders y is determined by the number of actual arrivals, so in the predictive system y is uniquely determined. Thus what actually determines stability is how close these prediction systems are to the real system. Using the core idea of SMPC [3], the form of Performance Index J is modified to correlate it with the probability of a random disturbance, namely the number of people, and ultimately the maximum possible prediction system is chosen, so that stable predictions can be produced. The specific algorithm is summarised below:

First: a system that randomly generates n actual 52-day arrivals, calculates their respective probability distributions, and selects the one with the smallest probability distribution in the sum of the differences of the actual number of arrivals.

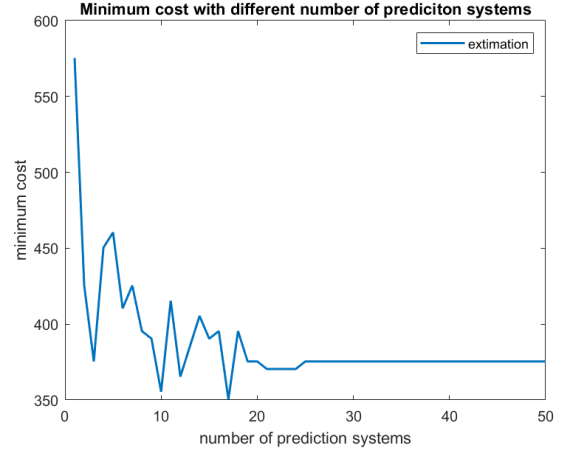
Second: Set up m forecasting systems at moment k_0 , each of which implements an optimal order allocation, namely it tries to satisfy remaining stocks of 0. If the actual system gets that the situation at the moment is that there is no need to order, its predictive order according to the probability of arrivals.

Third: The obtained m kinds of prediction systems are subjected to performance judgement. The judgement condition is to take the maximum of the sum of their probabilities,

namely to select the best ordering of the predicted system that is most likely to be the real system.

Fourth: Bring in the real system. Then, repeat the above process.

The following figure shows a plot of the prediction results obtained by applying SMPC at a prediction horizon of 2 days and a control horizon of 1 day:



As can be seen from the above figure, as the number of prediction systems increases, the final minimum cost gradually stabilises at around 375, namely a more stable prediction is eventually obtained.

III. CONCLUSION

In this paper, the minimum cost in the dynamic system is obtained by RHC, and an auto-adjusting variable-length control horizon and prediction horizon system is designed, and the effects of different lengths of control and prediction horizons on the results are observed. Finally stable prediction results are obtained by modifying the system to SMPC. However, since the performance index depends on the probability distribution of the number of people, the number of occurrences of 2 and 3 in the variable order y is larger than initially imagined, which is because although the most probable prediction system can be filtered out by this method, it still does not fully represent the actual system, and thus the model still has a lot of space for improvement.

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